

T05 — Density peaks at 4σ (phenix.find_peaks_holes)

Oracle: 23 + / 8 – peaks · agent listed 9 · 5 peaks above 5σ are not in agent’s table

Oracle σ	Nearest atom	Notes	In agent’s table?
+6.18	Glu A 54 CD	highest positive peak	no
+5.94	HOH S 34	near modelled water	no
+5.83	Thr A 67 CG2	side-chain density unaccounted	no
+5.61	Asn A 39 CB	matches agent’s 4.82σ near OD1	yes
+4.96	HOH S 736	matches agent’s 4.98σ entry	yes
−4.11	HOH S 401	modelled water with no density (negative peak)	no

Density-peak inventory is incomplete — fails the success criterion

Agent’s success criterion: **no unexplained $\Delta\rho > 4\sigma$** · agent reports 9 peaks, max 4.98σ.

Oracle: **23 + / 8 –** peaks at 4σ · max **+6.18σ** · 5 peaks above 5σ not surfaced.

Likely cause: agent’s peak-finding cutoff or merging step is more aggressive than the catalog oracle’s.

ASSUM_find_peaks_holes_cutoff_and_merging — known_violation.

Recommended remedy: re-run phenix.find_peaks_holes with map_cutoff=4.0 and accept default merging.